

Countable Borel Relations, Invariant Measures, and Graphings

First, we recall some definitions.

Definition:

- A relation R on a set X is a set of ordered pairs $R \subseteq X^2$. We will write xRy to denote $(x, y) \in R$. We also define the *sections*

$$R_x = \{y : (x, y) \in R\}$$

$$R^y = \{x : (x, y) \in R\}$$

and the inverse relation

$$R^{-1} = \{(y, x) : (x, y) \in R\}.$$

- We will let $\text{dom}(f)$, $\text{ran}(f)$, and $\text{graph}(f)$ denote the domain, range, and graph of f respectively.
- A graph G with vertex set X is a non-reflexive, symmetric relation on X . The neighbors of $x \in X$ in the graph G are the set of all $y \in X$ such that $(x, y) \in G$. The cardinality of the set of neighbors of x is called the degree of x , denoted $d_G(x)$.
- A path in G is a finite sequence of vertices $(x_0, x_1, \dots, x_n)$ where $(x_i, x_{i+1}) \in G$, and $x_i \neq x_j$ whenever $i \neq j$ except for possibly $x_n = x_0$. The equivalence classes of the relation xEy if and only if there exists a path from x to y defines the connected components of the graph. If a graph has one connected component, we say the graph is connected.
- If $x_n = x_0$ with $n \geq 3$, then we call the path a cycle. A graph that has no cycles is called acyclic, and a connected acyclic graph is called a tree. A rooted tree is a tree with a distinguished vertex x_0 called its root.
- If X is a standard Borel space and E is a Borel equivalence relation on X , then we say the relation is countable if $[x]_E$ is countable for each $x \in X$.
- If $\Gamma \curvearrowright X$ is a countable group acting on X such that Γ preserves the Borel structure of X , then we define the orbit equivalence relation to be $xE_\Gamma^X y$ if and only if there exists g such that $g \cdot x = y$.
- The orbit space X/Γ is defined to be the quotient X/E_Γ^X .
- If E is a countable Borel relation, then the *full group* of E , denoted $[E]$, is the set of Borel automorphisms whose graphs are contained in E . The set of partial Borel automorphisms is denoted $[[E]]$; that is,

$$[[E]] = \{\phi : A \rightarrow B : \phi \text{ is a bijection, } A, B \subseteq X \text{ are Borel, } \text{graph}(\phi) \subseteq E\}.$$

- We call a countable equivalence relation E on X *aperiodic* if every equivalence class $[x]_E$ is infinite.

Example: If X^Γ denote the space of functions of Γ onto X , then the left shift action of Γ on X^Γ is defined by $\gamma \cdot p(\delta) = p(\gamma^{-1}\delta)$.

If G is a standard Borel group and $\Gamma \subseteq G$ is a countable discrete subgroup (also known as a lattice), then the action of Γ onto G by left translation, taking $\gamma \cdot g = \gamma g$, yields the quotient space G/Γ given by the set of *right* cosets of Γ .

Theorem (Feldman–Moore): Let E be a countable Borel equivalence relation on X . Then, there is a countable group Γ and a Borel action of Γ on X such that $E_\Gamma^X = E$.

References

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